

Martin Wu

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Education

University of British Columbia

Bachelor of Applied Science in Engineering

Sep 2025 – 2029

Vancouver, BC

Technical Skills

Languages: C, C++, Go, Python, TypeScript, JavaScript

Embedded: STM32, FreeRTOS, SPI, CAN bus, HAL, CMake

Software: gRPC, React, Git, Docker, Vite, Playwright

Experience

Co-Founder & Software Engineer

Oct 2025 – Present

Supercluster | [GitHub](#)

Vancouver, BC

- Building a two-plane Go runtime for agent orchestration: control plane dispatches over mTLS gRPC (bidi `Connect` + unary `ReportCompletion`); data plane is the only path to backends
- Wrote `data-plane-local`: in-memory probe of the production 11-step dispatcher — identity gate, idempotent writes, intent-before-adapter audit, fail-closed denials; `go run`, no Docker
- Shipped the operator console (HTML/CSS/JS): live INTENT/RESULT/DENIED ledger, DAG workflows with fail-closed halt, 11 adapter sources, policy denials
- Planes share only `proto/agent`; every agent read/write is mediated by policy and a tamper-evident audit chain (metadata only — row values never cross)

Embedded Systems Engineer

Sep 2025 – Present

UBC Solar

Vancouver, BC

- Brought up STM32 Motor Drive Interface firmware (SPI, CAN, FreeRTOS) for Cascadia; 6 merged PRs on `firmware_v4`
- Wrote MAX31865 PT1000 driver with hardware-verified $<1^{\circ}\text{C}$ reads; faulted samples are debounced and withheld from CAN
- Implemented MDI CAN motor-command parsing from the driver display, with timeout-aware command caching for the motor controller
- Shipped steering-wheel firmware: cyclic drive-control CAN (regen, cruise, lights) and AS1115 hex speed display
- Wrote DRD external-lights firmware for the v4 vehicle

Projects

Songless | *React, TypeScript, Vite, Cloudflare R2, Web Audio* | Live

2026

- Song-intro guessing game with 100k+ visits; stage-locked Web Audio clues (0.01–15s) and R2-hosted full-track streaming
- Media pipeline: YouTube source audit, silence-trimmed encodes, storage-capped R2 uploads, 120-song catalogue

Us | *Phaser 4, TypeScript, Vite, Playwright*

2026

- Browser story game with a custom engine: 32px grid movement, save/fast-travel, dialogue, audio buses, 15+ scenes
- Playwright visual-audit gates on lighting, walk cycles, and collision so scenes cannot silently regress

Useful Tool Hub | *JavaScript, Python, Vercel* | Live

2025 – 2026

- Ten in-browser tools (convert, PDF, QR, image edit); YouTube/Instagram downloaders on Vercel Python, rest on-device

Awards

Mathematics (2023–2025): CSMC Top 0.5% nationally (71st) | AIME Qualifier 2x | Euclid Top 2% (85/100)